Minimax Estimation of distance on a surface and minimax manifold learning in the isometric-to-convex setting

Pierre Cornilleau
Abel Douzal
Ruben Ifrah

Dauphine PSL – ENS-PSL – École polytechnique

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2. Main result
3. Experiments
   - Implemented algorithms
   - Geodesics: robustness and accuracy
4. The Offset Method & Robustness Evaluation
   - Offset method: idea and guarantees
   - Robustness experiment near the cut locus
5. Perspective and conclusion

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Main goals

Setting: given a finite set of points $\{x_i\}_{1\leqslant i\leqslant n}$ on $\mathcal{M}$ a $\mathscr{C}^2$ d-submanifold,

- Non parametric estimation of intrinsic distances $d_{\mathcal{M}}(x_i, x_j)$
- Application in Manifold learning (i.e. dim reduction)
- Main non-parametric hypothesis :

$$\max_{x\in\mathcal{M}} \min_{1\leqslant i\leqslant n} \|x - x_i\| \leqslant \varepsilon$$

- Anterior bound for $\mathscr{C}^2$ submanifolds gives a relative error rate in $O(\varepsilon^{2/3})$:

$$\left| d_{\mathcal{M}}(x_i, x_j) - \widehat{d}_{ij} \right| \leqslant C\varepsilon^{2/3} d_{\mathcal{M}}(x_i, x_j)$$

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Main result (Arias-Castro & Chau)

Nota : for n points on $\mathcal{M}$, we have at best $\varepsilon \asymp ((\log n)/n)^{1/d}$.

Main results

1. **A Minimax Optimal Rate:** They prove that the theoretical optimal error rate for geodesic estimation is exactly quadratic in the sampling density $O(\varepsilon^2)$:

$$\left| d_{\mathcal{M}}(x_i, x_j) - \widehat{d}_{ij} \right| \leqslant C\varepsilon^2 \min(d_{\mathcal{M}}(x_i, x_j), \widehat{d}_{ij})$$

2. **Constructive Attainment:** This minimax rate cannot be attained by 1D graphs, but it *is* attained by geodesic paths calculated across the flat faces of a reconstructed mesh (Tangential Delaunay Complex).

Proof global structure for minimax bound in two parts

1. Upper bound (non constructive):
Prove $|d_{\mathcal{M}}(x,y) - \widehat{d}(x,y)|/d_{\mathcal{M}}(x,y) \lesssim \varepsilon^2$ using:

- Axiom of choice,
- Geodesic stability on nearly-isometric surfaces.

2. Lower bound (information-theoretic):

- Construct two manifolds with geodesic distances differing by $O(\varepsilon^2)$ that are statistically indistinguishable from n samples.
- No estimator can achieve better than $O(\varepsilon^2)$ worst-case error.

Key limitation: The ε^2 distortion is fundamental—intrinsic to the sampling resolution, not the algorithm.

Tangential Delaunay Complex (TDC)

Motivation: Need a provably good mesh reconstruction of $\mathcal{M}$ from point samples to compute geodesics.

Why TDC? Only known method simultaneously guaranteeing:

1. **Correct topology:** TDC is homeomorphic to $\mathcal{M}$ when $\varepsilon < \text{reach}(\mathcal{M})/2$.
2. **Geometric fidelity:** Hausdorff distance $d_H(\text{TDC}, \mathcal{M}) = O(\varepsilon^2)$.
3. **Geodesic stability:** Polyhedral geodesics on TDC approximate $d_{\mathcal{M}}$ with relative error $O(\varepsilon^2)$.

Proof Scheme: Attaining the Minimax Rate (1/2)

Goal: Prove that distances computed on the estimated TDC $\widehat{\mathscr{T}_c}$ satisfy the optimal bound:

$$\left|d_{\mathcal{M}}(x_i, x_j) - d_{\widehat{\mathscr{T}_c}}(x_i, x_j)\right| \lesssim \varepsilon^2 \min\Big(d_{\mathcal{M}}(x_i, x_j),\, d_{\widehat{\mathscr{T}_c}}(x_i, x_j)\Big).$$

Proof Steps:

1. **Tangent Space Estimation (Prop. 4.4):** Local PCA yields estimates $\widehat{T}_i$ with bounded angle error: $\angle\big(T_{x_i}\mathcal{M}, \widehat{T}_i\big) \lesssim \varepsilon$.
   - This is achieved by analyzing the local covariance matrix of points within a neighborhood of radius proportional to ε.

2. **Intermediate "Phantom" Surface $\widetilde{\mathcal{M}}$ (Prop. 4.6):** Because $\angle(T_i, \widehat{T}_i)$ is small, there exists an intermediate smooth surface $\widetilde{\mathcal{M}}$ that:
   - Interpolates the samples x_i and has tangent spaces *exactly* equal to $\widehat{T}_i$.
   - Retains a strictly positive reach, ensuring it is a well-behaved manifold.
   - Is ε^2-close to $\mathcal{M}$ in Hausdorff distance, meaning the metric distortion is controlled: $|d_{\mathcal{M}} - d_{\widetilde{\mathcal{M}}}| \lesssim \varepsilon^2 d_{\mathcal{M}}$.

Proof Scheme: Attaining the Minimax Rate (2/2)

3. **TDC Distortion (Thm. 4.3):** Since $\widehat{\mathscr{T}}_c$ is built using the *exact* tangent spaces of $\widetilde{\mathcal{M}}$, theoretical TDC guarantees fully apply.
   - $\widehat{\mathscr{T}}_c$ is an $O(\varepsilon^2)$-distortion map of $\widetilde{\mathcal{M}}$.
   - This yields a direct bound on the distances: $\left|d_{\widetilde{\mathcal{M}}} - d_{\widehat{\mathscr{T}}_c}\right| \lesssim \varepsilon^2 d_{\widehat{\mathscr{T}}_c}$.

4. **Conclusion via Triangular Inequality:** Adding the metric error between the true and phantom surfaces yields the final bound:

$$\left|d_{\mathcal{M}} - d_{\widehat{\mathscr{T}}_c}\right| \leq \left|d_{\mathcal{M}} - d_{\widetilde{\mathcal{M}}}\right| + \left|d_{\widetilde{\mathcal{M}}} - d_{\widehat{\mathscr{T}}_c}\right| \lesssim \varepsilon^2 \left(d_{\mathcal{M}} + d_{\widehat{\mathscr{T}}_c}\right)$$

   - For ε chosen small enough (e.g., ensuring the constant $C\varepsilon^2 \leq 1/3$), this algebraic sum naturally forces the final bound to take the required min formulation.
   - This perfectly bridges the gap between the true, unknown manifold $\mathcal{M}$ and the computable mesh $\widehat{\mathscr{T}}_c$.

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From Isomap to Tangential Delaunay Complex

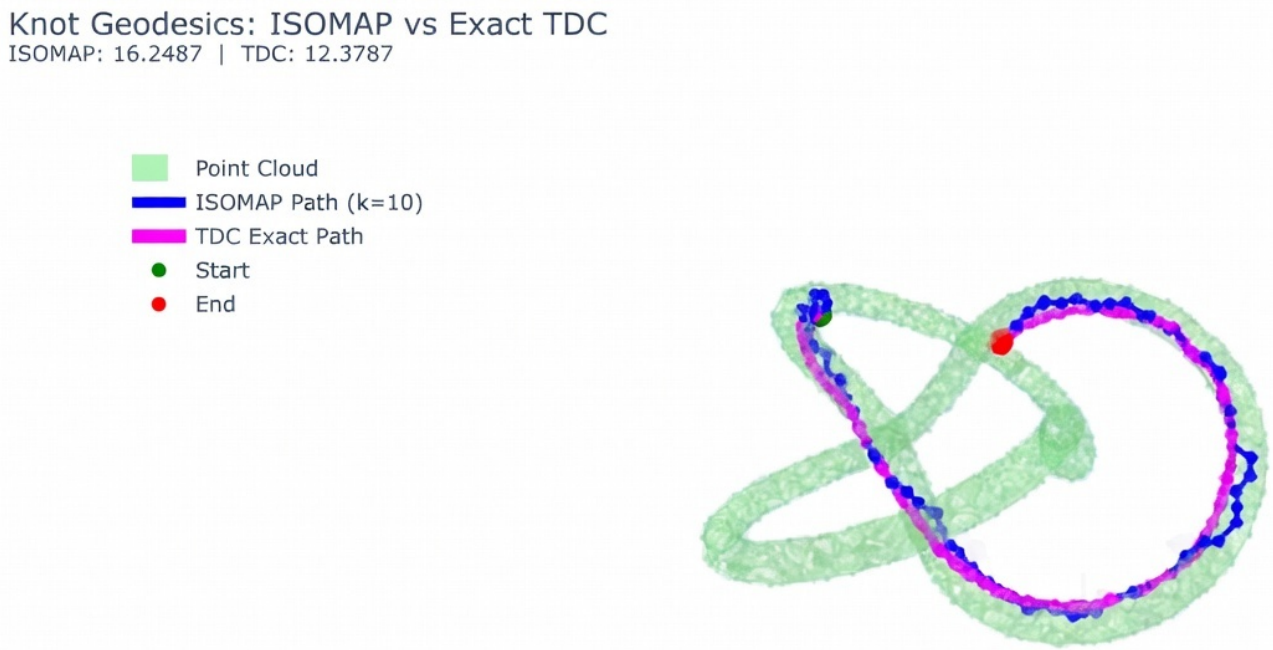
We compare two ways to approximate geodesic distances on manifolds:

- **Graph-Isomap (file `ISOMAP.py`):**
  - Build a k-nearest neighbor graph with edge weights $\|x_i - x_j\|$ in
  - Compute all-pairs shortest–path distances on this graph as geodesic
  - Optionally embed via classical MDS (Isomap algorithm).
- **Tangential Delaunay Complex (file `TDC.py`):**
  - Reconstruct a mesh of the manifold using the tangential Delaunay complex.
  - Compute geodesic distances on this mesh.

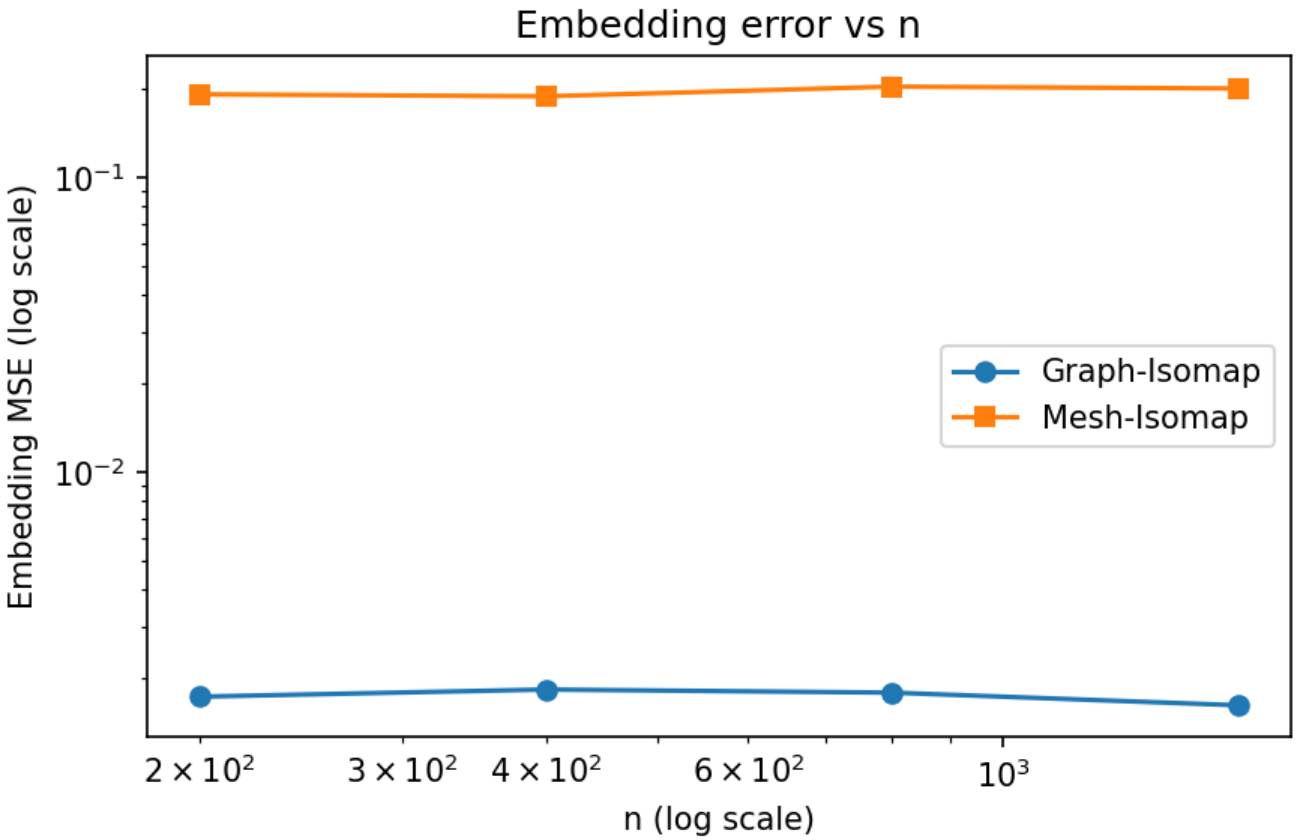
This mirrors the theory: first reconstruct a good mesh (via TDC), then compute geodesic

Experiments – geodesic paths (knot)

- Compare D_{graph} (Isomap k-NN path) with the exact TDC geodesic on a tubular knot.
- Visually: TDC (magenta) hugs the true tube; the graph path (blue) sometimes shortcuts through the ambient space.
- Numerically: we report both path lengths in the title.

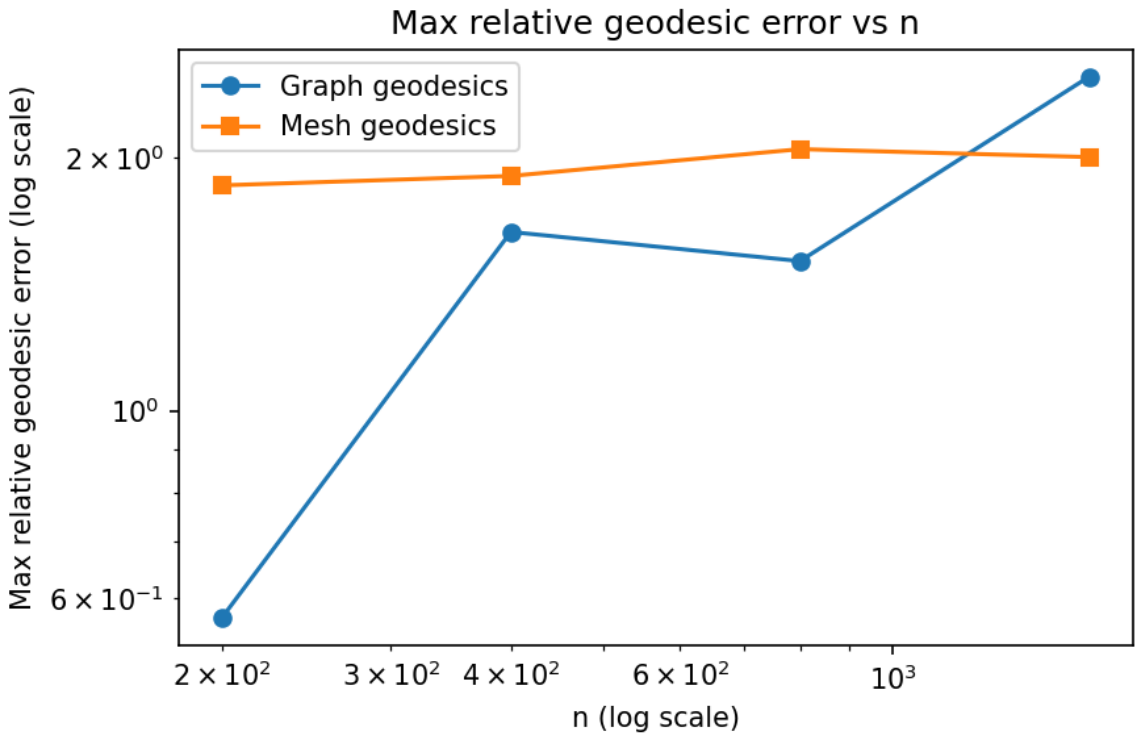


Embedding error vs n



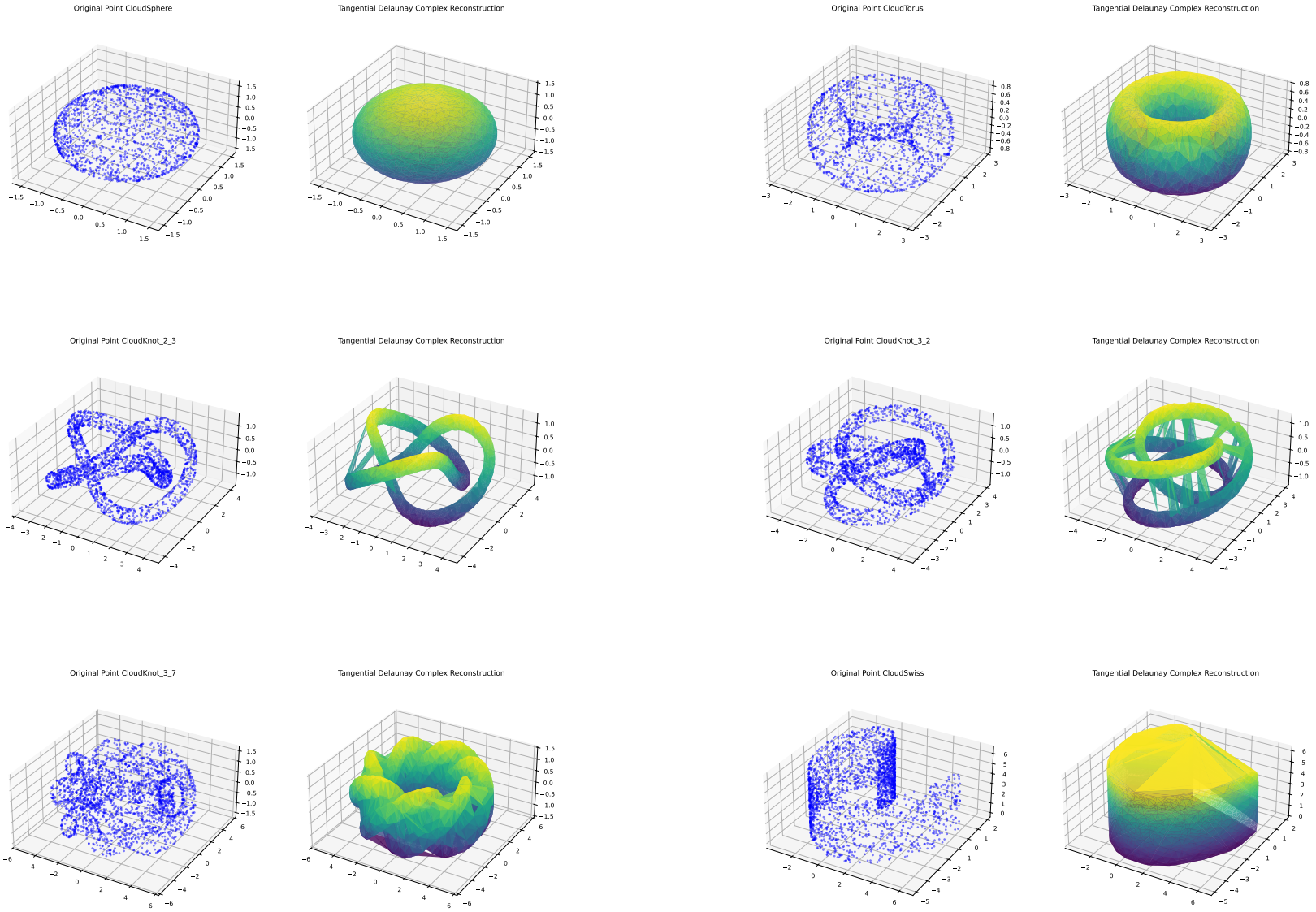
- Graph-Isomap achieves a much smaller embedding MSE than our naive Mesh-Isomap on this simple surface.
- This shows that in this toy setting, the k-NN graph already approximates the latent metric very well, while our mesh construction is not optimized.
- The result is compatible with the paper: we are *not* implementing their tangential Delaunay complex, only a crude proxy.

Geodesic error and minimax link

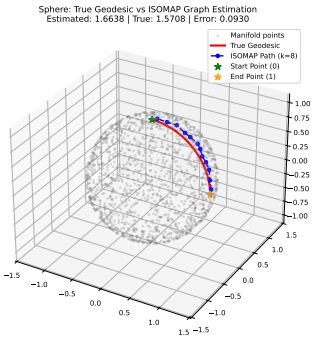


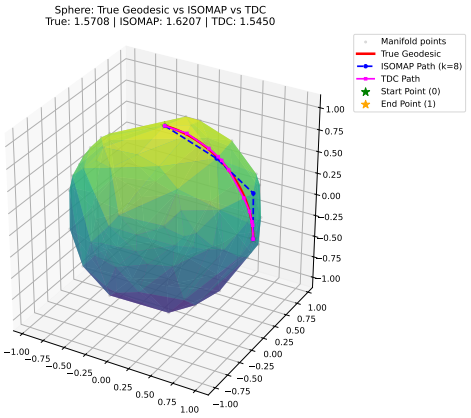
- Both methods have max relative geodesic error of order 1–2; curves cross and there is no clear dominance in finite samples.
- Arias-Castro & Chau prove minimax upper and lower bounds for $\sup_{x,y} |d_{\mathcal{M}}(x,y) - \widehat{d_n}(x,y)|$ when the surface is reconstructed by a tangential Delaunay mesh.
- Our plots are only a qualitative illustration: they show the importance of geodesic error for embedding quality, not a numerical verification of the exact minimax rate.

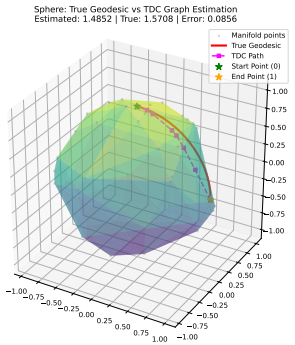
Manifold reconstructions using the Tangential Delaunay Complex

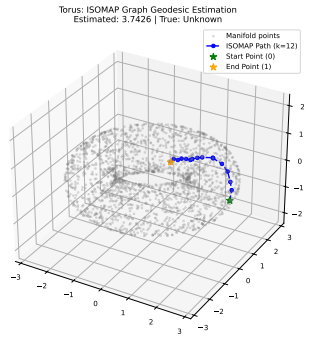


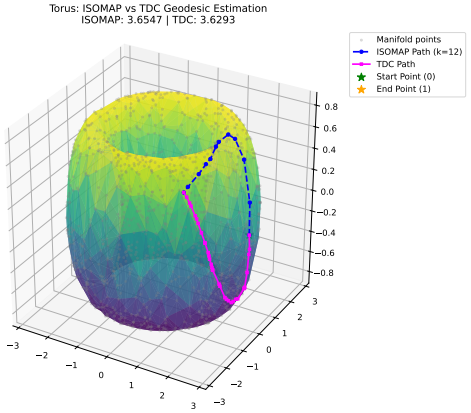
Geodesic reconstructions (TDC and ISOMAP)

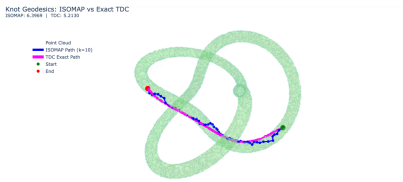












Geodesic quality: the minimax rate

Setting: $\mathcal{M}$ a k-manifold embedded in $\mathbb{R}^p$, sampled with n points.

Mesh resolution: The covering radius

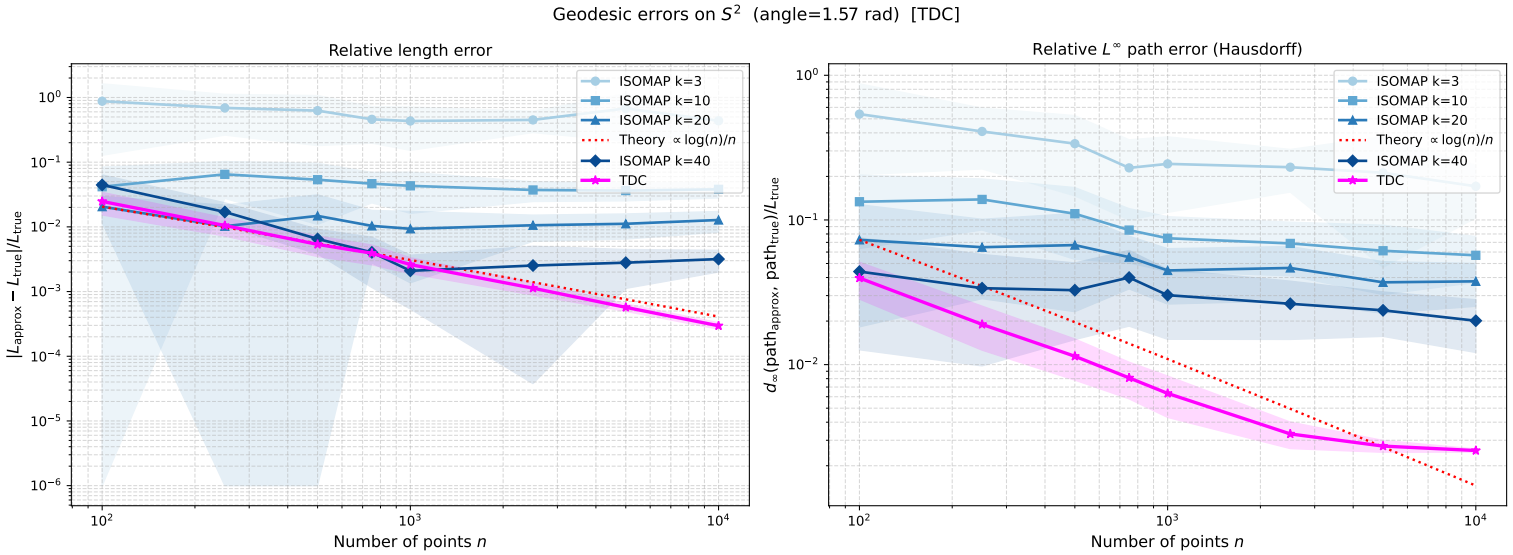
$$\varepsilon := \max_{x \in \mathcal{M}} \min_{i=1,\dots,n} \|x - x_i\| \asymp \left(\frac{\log n}{n}\right)^{1/k}.$$

Use of the minimax result proven : In the isometric-to-convex setting, the relative geodesic error satisfies

$$\sup_{x_i \neq x_j} \frac{\left| d_{\mathcal{M}}(x_i, x_j) - \widehat{d}_{i,j} \right|}{d_{\mathcal{M}}(x_i, x_j)} \lesssim \varepsilon^2 \asymp \frac{\log n}{n} \quad (\text{for } k = 2),$$

and *no estimator* can improve this worst-case rate. TDC-based geodesics attain it.

Empirical validation on the sphere ($k = 2$)



Our experiments confirm:

- TDC tracks the $O(\log n/n)$ minimax rate closely.
- Graph-Isomap performs well but with larger constants.
- Both methods converge as n increases, consistent with theory.

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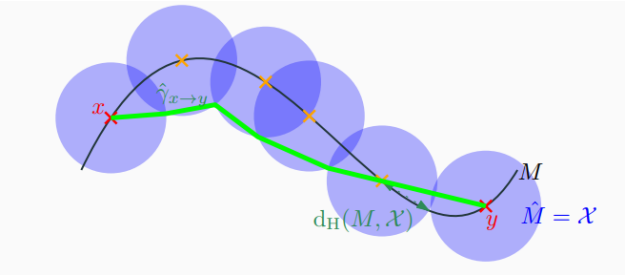
A Volumetric Approach: Geodesics of Offsets

A critique of TDC:

- While the Tangential Delaunay Complex achieves minimax optimality ($O(\varepsilon^2)$), zero-thickness reconstructed surfaces are topologically unstable and computationally heavy in high ambient dimensions.

Offset Method (from class on Geometric Inference & *Aamari et al. - 2023*):

- Instead of forcing the path to travel exactly on a 1D skeleton or a 2D flat face, the path float within a volumetric bound



- Compute an estimator $\widehat{M} \subset \mathbb{R}^p$ such that $d_H(M, \widehat{M}) \leqslant \varepsilon < rch_M/2$.
- Define the ε-offset: $(\widehat{M})^\varepsilon := \{u \in \mathbb{R}^p \mid d(u, \widehat{M}) \leqslant \varepsilon\}$.
- The geodesic $d_{(\widehat{M})^\varepsilon}(x, y)$ is calculated as the shortest continuous path floating entirely within this thickened offset volume.
- $\sup_{x \neq y \in M} |d_M(x, y) - d_{(\widehat{M})^\varepsilon}(x, y)| \leqslant \varepsilon$

Experimental Setup: Robustness on the Sphere

The Setup:

- **Manifold:** 3D Sphere, $N = 1500$ points.
- **Start point:** North Pole.
- **End point:** Polar angle α varying from 155° to 179°

Topological Instability:

- As the target approaches the antipode (the cut locus at 180°), the opposing path becomes nearly identical in length.
- Discrete estimators are highly susceptible to snapping to the "wrong" hemisphere due to sampling density variations.

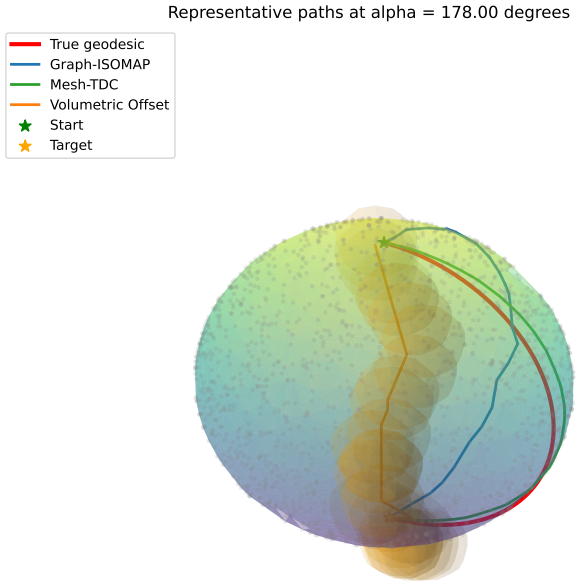
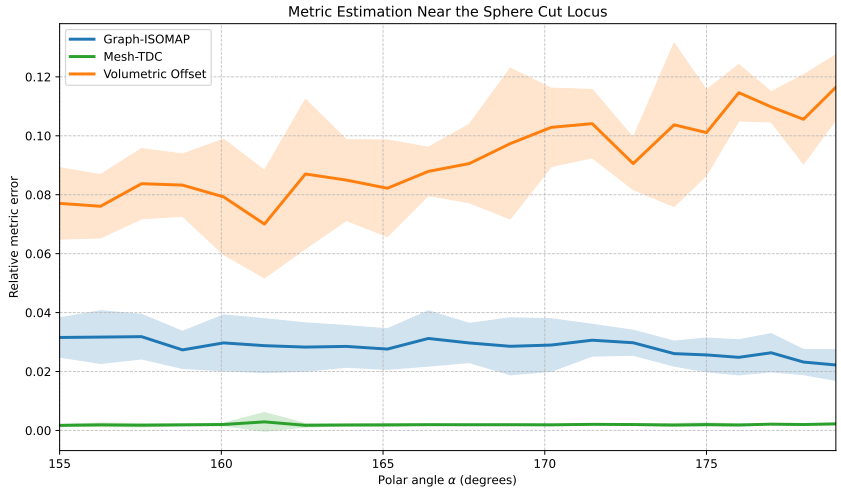


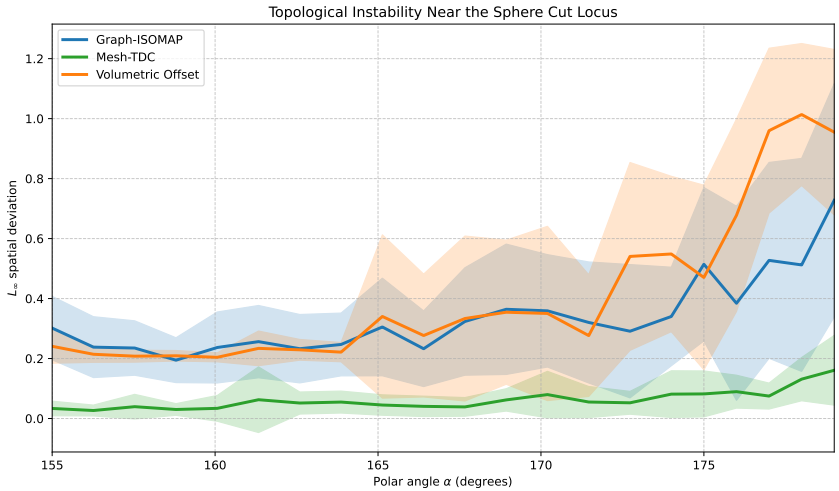
Figure: geodesic estimation for an angle of 179° on the sphere (1500 points)

Results & Commentary: Robustness Comparison

Relative Metric Error



Spatial Deviation (L_∞)



Observations:

- **Mesh-TDC :** empirically consistent with the minimax optimal $O(\varepsilon^2)$ bound of Arias-Castro & Chau: metric error is smallest and the path stays glued to the true geodesic ($L_\infty \approx 0$).
- **Graph-ISOMAP:** systematic baseline metric error and mild spatial deviation, illustrating the "zig-zag" tax inherent to 1D discrete graph edges.
- **Volumetric Offset ($d_{\widehat{(M)}^\varepsilon}$):** highest error and spatial wandering.
  Note: difficult implementation of the method using discrete voxel-grid pathfinding, leading to flawed geodesic estimation

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Perspective and conclusion

What we achieved: Validated theory numerically: TDC geodesics exhibit optimal minimax rate for 2-manifolds in $\mathbb{R}^3$ and superior robustness than Isomap.

Open questions & limitations:

Scaling in high dimension Does algorithms for computing geodesics scale well ?

Noisy data: How robust is the ε^2 rate when points are sampled with small normal noise?

Computational cost: Can other mesh constructions achieve optimal rates more efficiently?limits ?

General settings: Extensions beyond isometric-to-convex manifolds? Verification of reach assumptions in real data?

Thanks for your attention!